

Performance Analysis of Various Techniques Applied in Human Identification Using Dental X-Rays

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Abstract This paper proposes a novel approach for automating the analysis of identifying the person based on their ante mortem and postmortem reports. This approach involves three techniques (i.e.) morphological contour detector, Gaussian filtering and an existing semi-automatic contour extraction method. Forensic dentistry involves the identification of people based on their dental records, mainly available as radiograph images. Our goal is to automate this process using image processing and pattern recognition techniques. Given a postmortem radiograph, we search a database of antemortem radiographs in order to retrieve the closest match with respect to some salient features. In this paper, we use the contours of the teeth as the feature for matching. The algorithm completes the task in three steps: radiograph segmentation, pixel classification and contour matching. In this paper a hit rate of 0.7 is achieved by the Morphological contour detectors which are comparable with the other two methods.

Keywords Semi automatic contour extraction · Morphological contour detection · Gaussian filtering · Hit rate · Matching distance

Introduction

The objective of the work here is to automate the process of forensic dentistry. The main purpose of forensic dentistry is to identify deceased individuals, for whom other cues of biometric identification (e.g., fingerprint, face, etc. [1, 2]) may not be available. On the basis of the time of Acquisition, there are two classes of dental radiographs. The radiographs acquired after the death are called the Post-mortem (PM) radiographs, and the radiographs acquired while the person is alive are called the Ante-mortem (AM) radiographs (Fig. 1). The AM radiographs, collected in the dentists' office, are labeled with patients' names. The method used in dental biometrics is matching the unlabelled PM radiographs against the database of labeled AM radiographs. If the set of teeth in a PM radiograph sufficiently matches the teeth in an AM radiograph, the identity of the PM radiograph is obtained. A manual comparison between the AM and PM records is based on a systematic dental chart prepared by forensic experts [2, 3]. In this chart, a number of distinctive features are noted for each individual tooth. These features include properties of the teeth (e.g., tooth present/not present, crown and root morphology, pathology and dental restorations), periodontal tissue features, and anatomical features. Depending on the number of matches, the forensic expert rejects or confirms the tentative identity. There are several advantages for automating this procedure. First, an automatic process will be able to compare the PM records against AM records pertaining to multiple identities in order to determine the closest match. Second, while a manual (non-automated) system is useful for verification on a small data set, an automatic (or semi-automatic) system can perform identification on a large database.

For the automated identification, the dental records are usually available as radiographs (Fig. 1). An automated

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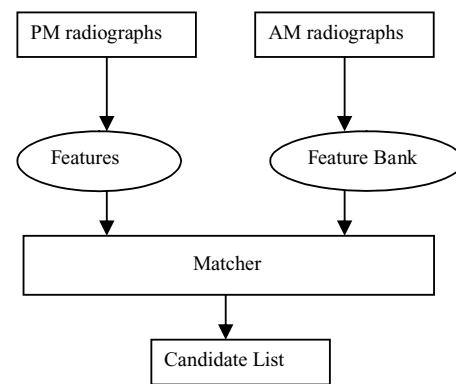
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Fig. 1 Dental radiograph

dental identification system consists of two main stages: feature extraction and feature matching. In this paper, the feature extracted is the tooth contours because they remain more invariant over time compared to some other features of the teeth. In earlier methods, the manual selection of ROI was counter to our goal of automated identification.

In this paper three techniques like Morphological contour detector [5], Gaussian filtering [6] and an existing semi-automated dental identification [2] system are discussed. This paper presents a new operator for contour detection. The morphological contour detector operator uses a variant of the morphological closing operator, which we have called asymmetrical closing. It consists of the successive application of different morphological transformations using different structuring elements. Each of these structuring elements used to probe the image under study is tuned to affect contour of different orientation and brightness. We found that this kind of approach, based on brightness comparisons, leads to better quality results than others and is achieved at a lower computational cost.

Gaussian filter is widely used for image smoothing but it is well known that this type of filters blur the image features (e.g., edges). The extension of Gaussian filters is the anisotropic filtering (bilateral filtering or PDE-based anisotropic diffusion) for feature-preserving smoothing. Noise is commonly seen in many types of images (for example, biomedical images and remote sensing images). It is in a great demand to smooth the images before other tasks could be conducted. Gaussian low pass filtering is known to be an efficient and simple way for image smoothing [6]. However, it is well known that Gaussian filtering blurs the image edges while smoothing image noise. The reason is that Gaussian filters are isotropic in the sense that all surrounding pixels affect the center pixel in a similar fashion regardless their intensity variations. Hence, the edges and the noise are treated in the same way, which yields noise reduction as well as edge blurring. To remedy the problem of traditional Gaussian filtering, people have proposed lots of methods, trying to achieve the goal of feature-preserving smoothing. All of these methods are called anisotropic filtering. In this paper this filtering is used as another method

**Fig. 2** Logical diagram of dental biometric system

for tooth contour extraction. The logical diagram of the dental Biometric system is shown in Fig. 2.

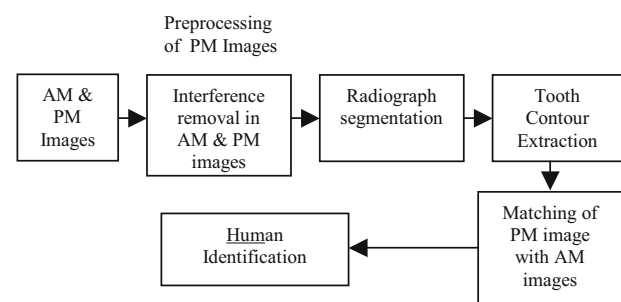
In this paper, the segmentation algorithm is developed for the detection of ROI. Further, a probabilistic method is introduced to automatically find the contours of teeth. However, a fully automatic feature extraction method is still not capable of handling the large variance in image quality and the appearance of teeth. Thus, human intervention is needed to initialize certain algorithmic parameters and correct errors in some problematic images.

Methodology

The three main stages of this technique are Radiograph segmentation, feature extraction and feature matching. The processing flow of this method is shown in Fig. 3. In this diagram PM refers to postmortem and AM refers to antemortem image. The goal of radiograph segmentation is to segment the Radiograph into blocks such that each block has a single tooth in it. This helps us to define the ROI associated with every tooth.

Radiograph segmentation

The goal of the segmentation is, to define the Region of interest (ROI) that represents exactly one tooth. The mathematical

**Fig. 3** The processing flow

morphological operation and Gaussian filtering techniques are mainly used in order to obtain the contour of the tooth from the desired ROI. Prior to the radiograph segmentation, the input images are preprocessed to remove the noise. The objective of using median filter is to achieve noise reduction rather than image blurring. Here the gray level of each pixel is replaced by the median of the gray level in a neighborhood of that pixel. This filtering operation is particularly effective when the noise pattern consists of strong, spike like components and the characteristic to be preserved is edge sharpness.

Gap valley detection

The teeth usually have a higher gray level intensity than the jaws and the other tissues in the radiographs. Thus the gray level intensity can be considered as a cue for us to separate the upper and lower jaws for which we construct the y -axis projection histogram by summing up the intensities of pixels along each row parallel to the x -axis. The gap between the jaws has low intensity values, which is useful for creating a valley in the projection histogram, which is called as the gap valley [2]. But, there may occur many other valleys in the projection histogram which demands the user assistance in detecting the gap valley.

The position of the user estimation of the gap valley is initialized to be $\hat{y}$ and the other valleys obtained from the projection histogram are represented by $v_i, i = 1, 2, 3, \dots, m$. Each v_i is associated with D_i and y_i where D_i is the intensity obtained at the gap valley and y_i is the position of v_i (Fig. 4a). To obtain the actual gap valley, maximize the probability function representing the gap valley denoted by $p_{v_i}(D_i, y_i)$. The depth and position of the valley are two independent variables, write the probability function as

$$p_{v_i}(D_i, y_i) = p_{v_i}(D_i)p_{v_i}(y_i) \quad (1)$$

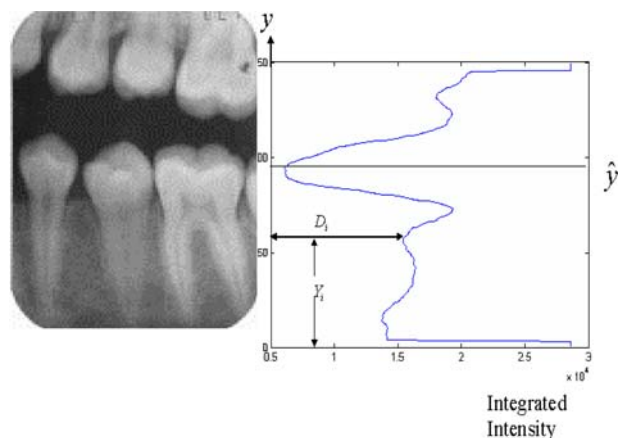


Fig. 4a Integral projection on the y -axis.

where

$$p_{v_i}(D_i) = c \left(1 - \frac{D_i}{\max_k D_k} \right) \quad (2)$$

$$p_{v_i}(y_i) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(y_i - \hat{y})^2/\sigma^2} \quad (3)$$

The ‘ c ’ in eqn. (2) is a constant and is used to ensure that the summation of the probability mass function yields 1. The way the gray level profile acts as cue is simply now restated as lower the intensity, larger is the probability that it is the gap valley and is represented in eqn. (2). The user initialization need not always point to the exact position of the gap valley and this error measure is incorporated in eqn. (3) as the Gaussian distribution. The Gaussian distribution models the required fact that lower the error, larger is the probability of the $\hat{y}$ being the gap valley. The true gap valley is now the maximization of eqn. (1) yielding v to be the gap valley as mentioned in eqn. (4).

$$p_v(\hat{D}, \hat{y}) = \max_i p_{v_i}(D_i, y_i) \quad (4)$$

The same gap valley technique is applied to separate the individual tooth in both upper and lower jaw in vertical direction and is shown in Fig. 4a and 4b.



Fig. 4b Integral projection on the X -axis

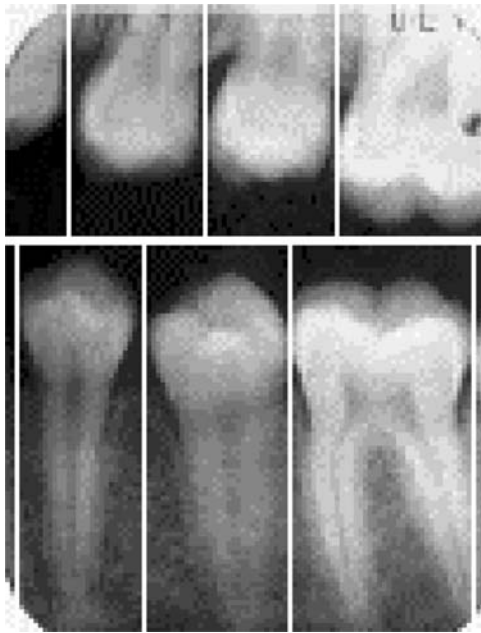


Fig. 5 Segmentation of tooth (place shifted)

Segmentation of tooth

Using the above gap valley technique, maximum probability value will be calculated (which indicates air gap valley), and the horizontal and vertical lines will be drawn in that positions to segment each tooth from neighbors in both upper and lower jaw and is shown in Fig. 5.

Shape extraction using morphological contour detector

Contours constitute attractive 2D features, often used in computer vision for tasks such as stereovision, 3D interpretation, motion estimation, and structure from motion [5]. Mathematical morphology is a methodology for image analysis that has been widely used in computer vision. The principle of all basic morphological operators is to probe the image under study with a structuring element. The shape of the structuring element defines a set $I_{SE}(x, y)$ that includes all pixels of the image hit by the structuring element. The structuring elements are shown in Fig. 6a–d. The erosion of an image I with a structuring element SE is given by:

$$I_{SE}^e(x, y) = \min I_{SE}(x, y) \quad (5)$$

Similarly, the dilation of an image I with a structuring element SE is given by:

$$I_{SE}^\delta(x, y) = \max I_{SE}(x, y) \quad (6)$$

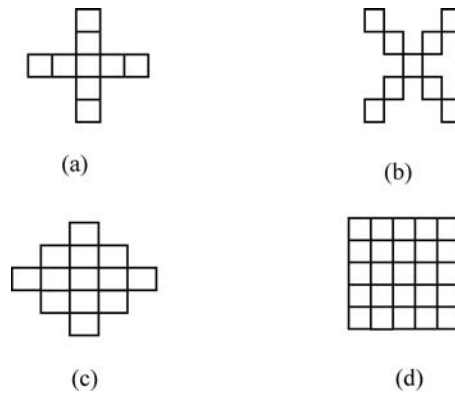


Fig. 6 Structuring elements

The opening of an image I by a structuring element SE is defined as an erosion followed by a dilation:

$$I_{SE}^o = (I_{SE}^e)_{SE}^\delta \quad (7)$$

where SE is the symmetrical transposition of SE with respect to its origin. The closing of an image I by a structuring element SE is defined as a dilation followed by an erosion:

$$I_{SE}^c = (I_{SE}^\delta)_{SE}^e \quad (8)$$

In the context of contour detection, one interesting choice is to consider a cross-shaped structuring element Fig. 6(a). Indeed, because of the particular shape of this structuring element, the opening and closing operators alter mainly this kind of image structure. Asymmetrical closing, that is, the dilation of an image using a given structuring element followed by an erosion using another structuring element (note that asymmetrical opening could also have been considered). The central idea is to make dilation and erosion complementary in terms of the type of contours they affect. This can be realized by choosing a cross as the first structuring element and a lozenge for the second one Fig. 6(c). The values used inside each square of structuring element are different for contour detection than the corner detection. We can then write the asymmetrical closing as:

$$I_{+, \diamond}^c = (I_+^\delta)_{\diamond}^e \quad (9)$$

$$c_+(I) = |I - I_{+, \diamond}^c|. \quad (10)$$

To detect the missing contour points, the following operator can be used (which is a 45 degree rotated version of the preceding one):

$$c_{\times}(I) = |I - I_{\times, \square}^c|. \quad (11)$$

Shape extraction using Gaussian filtering

A centre-surround mask can be used to locate grey-level boundaries in an image. It seems reasonable to combine the centre-surround operation with Gaussian smoothing to find the boundaries at different scales in the image. The Gaussian edge preserving and smoothening kernel along the X -direction is H_X is:

$$\frac{\text{EXP}\left(\frac{-X^2}{2\sigma^2}\right)}{\sigma\sqrt{2\pi}}(U, \sigma) \times \left(-X \times \frac{\text{EXP}\left(\frac{-X^2}{2\sigma^2}\right)}{\sigma^2}\right)(U, \sigma) \quad (12)$$

In this eqn. (12), where ‘ X ’ is the coordinate of the kernel mask with the size of 7×7 . ‘ σ ’ is the standard deviation. ‘ U ’ is parameter related to coordinate i, j , where $i, j = 1, 2, \dots, 7$. The kernel along Y -direction is H_Y where $H_Y = \tilde{H}_X$.

Using the eqn. (12), Gaussian mask with the size of 7×7 is generated for ‘ X ’ and ‘ Y ’ directions. The mask slid over the Dental radiograph image in horizontal and vertical directions will give the boundaries of the teeth.

Shape extraction using Semi automatic contour extraction method

A tooth has two main parts: the crown, which is above the gumline, and the root, which sits in the bone below the gum. Due to the overlap of the tooth root image with the image of the jaws, the root is not as visible as the crown in the radiographs due to the lower differential in tissue density [2, 8]. Thus the crown is identified first, followed by the root contour extraction.

Crown shape extraction

By drawing a line through the crown center, the ROI is divided vertically into two rectangles. The crown area has two classes of pixels: the toothpixels ω_t and the background pixels ω_b . The density function is:

$$P(I) = p(I|\omega_b)P(\omega_b) + p(I|\omega_t)P(\omega_t), \quad (13)$$

where, I is the pixel intensity, $P(I|\omega_b)$ is the distribution of intensities of background pixels and $p(I|\omega_t)$ is the distribution of intensities of teeth pixels. To detect the contour of the tooth crown, we perform a radial scan from the crown center C . Specifically, we draw a radial line from the crown center C . For each point P on the line, let P_{inner} and P_{outer} be the neighbors of P along the radial line. Then the probability for the point P to be a point on the contour of the current tooth

is defined as

$$p(E) = p(\omega_b | I_{\text{outer}}) \cdot p(\omega_t | I_{\text{inner}}), \quad (14)$$

Root shape extraction

Once the crown shape is extracted, we traverse from the two ends of the shape boundary to find the root boundary. The left end of the crown is set as the first point on the left contour of the root, and the right end of the crown is the first point of the right contour. We determine the position of each new contour point on the root boundary by the position of the previous contour point and its own context. As a measurement of the context, we define two attributes at each point: I_{inner} for the inner intensity of the contour and I_{outer} for the outer intensity. In other words, for the teeth in the lower jaws, if the i th point on the left/right root contour has coordinates $(x_i; y_i)$, the $(i + 1)$ th point, $(x_i + 1; y_i + 1)$, is computed iteratively as

$$\begin{aligned} x_i + 1 &= \arg \max_{x_i - r \leq x \leq x_i + r} (I_{\text{inner}} - I_{\text{outer}}), \\ y_i + 1 &= y_i + h, \end{aligned} \quad (15)$$

where r is the radius of the search space and h is the increment in the vertical position for each new point. Once we get the crown contour and the root contour, we connect them to form the contour of the whole tooth.

Matching distance calculation

The contours extracted from the query image must be matched to the contours extracted from the database images. Because the PM images are usually captured several years after the AM images are acquired, the shapes of the teeth could have changed due to teeth extraction or the growth of teeth [9, 10]. If we assume there are no such changes, the AM and PM radiographs differ only in terms of scaling, rotation, translation and the change of the imaging angle. Because of the criteria of the intraoral radiographic examinations [11], the change of imaging angle is not severe. So, currently, we omit this variation. Other differences can be factored out by a rigid transformation.

Affine transformation

An affine transformation is any transformation that preserves co linearity (i.e., all points lying on a line initially still lie on a line after transformation) and ratios of distances (e.g., the midpoint of a line segment remains the midpoint after transformation). In this sense, affine indicates a special class of projective transformations that do not move any objects

from the affine space $\mathbb{R}^3$ to the plane at infinity or conversely. An affine transformation is also called an affinity. A transformation changes the positions of points in the plane. A set of points, when transformed, may as a result acquire a different shape. Applying an affine transformation to a uniformly distorted image can correct for a range of perspective distortions by transforming the measurements from the ideal coordinates to those actually used. Determining the best contour alignment requires us to find the rigid transformation T , which minimizes the MD defined above. The rigid transformation, T , is a function of the form

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(\phi) = A\phi + \tau, \quad (16)$$

where, $\phi = (x, y)_t$ represents a point in the query shape, $T(\phi)$ is the result of applying the transformation T on ϕ , A is the transformation matrix, and τ is a translation vector. The parameters A and τ can be represented as

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \psi_x & 0 \\ 0 & \psi_y \end{pmatrix} \tau = \begin{pmatrix} \tau_x \\ \tau_y \end{pmatrix}.$$

For a given Query image we generate several sub images from the database image each containing the same number of teeth as the Query image [2, 8]. If the teeth in query image Q and database sub-images D are labeled as $1, 2, \dots, N$ from left to right, then

$$Q = \{\Gamma_i\}_{i=1}^N$$

and

$$\Gamma_i = \{\gamma_{i,j}\}_{j=1}^{N_{\Gamma_i}}$$

The matching distance MD is given by,

$$MD(T, Q, D) = \sum_{i=1}^N D(T, \Gamma_i, \phi_i), \quad (17)$$

where,

$$D(T, \Gamma_i, \phi_i) = \sum_{\phi_{i,j} \in \phi_i} \min_{\gamma \in \Gamma_i} \|T(\phi_{i,j}) - \gamma\|.$$

There are a total of 5 parameters in the transformation T , $\{\theta, \psi_x, \psi_y, \tau_x, \tau_y\}$, which are optimized by searching for the best alignment between the transformed query shape and the database shape. The parameters

$$\theta = \theta_d - \theta_q, \quad \psi_x = W_d/W_q, \quad \psi_y = \psi_x, \\ \tau_x = Cd_x - Cq_x, \quad \tau_y = Cd_y - Cq_y,$$

Fig. 7 Input image



where θ_q and θ_d are the orientations [2,4] of the tightest fitting rectangles for the query shapes and the database shapes, respectively, W_q and W_d are the widths of the rectangles, and $(Cq_x; Cq_y)$ and $(Cd_x; Cd_y)$ are the centers of the rectangles.

Results and discussion

The algorithm is developed using Matlab7. The dental image of size (108,115) has been taken as an input image as shown in Fig. 7.

The input image is segmented to isolate the tooth using gap valley detection technique and is shown in Fig. 8.

The root and crown shapes are extracted using the semi-automated method (i.e.) using the intensity of pixels and the shape is shown in Fig. 9. It is shown only for the lower jaw. In the same way the upper jaw can also be extracted.

The results of Morphological contour detector and the Gaussian filter are shown in Fig. 10.

The comparative result for AM and PM images with their matching distance for the three techniques are shown in Figs. 11, 12, 13.

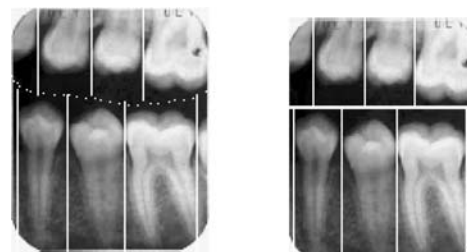


Fig. 8 Segmentation of tooth using Gap Valley detection method

Fig. 9 Shape extraction-Semi automatic contour Extraction method



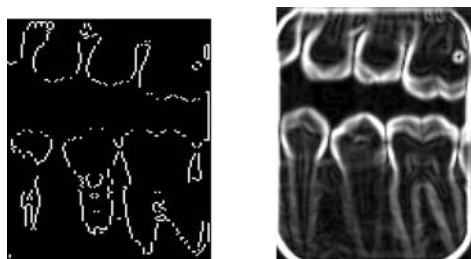


Fig. 10 Shape extraction using morphological contour detector and gaussian filtering

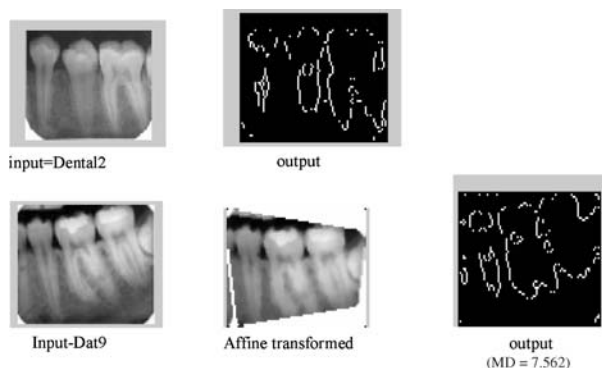


Fig. 11 Morphological contour detector

Once the shape has been extracted, the shape matching can be done by calculating the matching distance. The matching distance is calculated for the 15 Query images with the database of 30 images. Out of 15 Query images one image (Dental2) and its matching distance against several database images has been listed in the Table. 1 for all the three methods. From the table it is clear

Table 1 Comparison of database and MD

Database	Q = Dental2		
	Morphological contour detection	Gaussian filtering	Semi automatic contour extraction
Db3	9.2690	22.1562	40.1
Db21	11.9745	31.3295	71.1
Db2	7.7423	15.9641	67.46
Dat9	7.562*	14.593*	40.09*
Dat3	8.4813	20.235	52.93

Table 2 Parameter comparison

		Database image	Morphological contour detection	Gaussian filtering	Semi automatic contour extraction
Hit rate	Dental2	Dat9	0.73	0.6	0.67
Matching distance			7.562	14.593	40.09
Computation time (sec)			1.98	1.9	2
Storage capacity (KB)			2	1.5	2

that Dental2 has been matched with Dat9 only but with different Matching distance. The '*' denotes the best match.

The comparison results for the three methods are shown here. Out of 15 Query images, 11 of them were ranked first as Genuine images for the Morphological contours detector, 2 were among top 2 retrievals and 3 were among top 3 retrievals and correspondingly hit rate and ranking were calculated. In retrieval performance, 4 query images were not correctly matched due to (i) similarity of tooth with other individuals, (ii) incorrect contour detection, (iii) unavailability of proper region of jaw portion and (iv) poor quality of the image.

Comparatively, the minimum matching distance (MD) value was obtained in morphological operation, not in Gaussian and semi-automated techniques may be because of more number of pixels extracted within the tooth, other than the actual pixels in the contour. These points may also be useful to obtain the similarity of pulp region in the tooth between the AM and PM images.

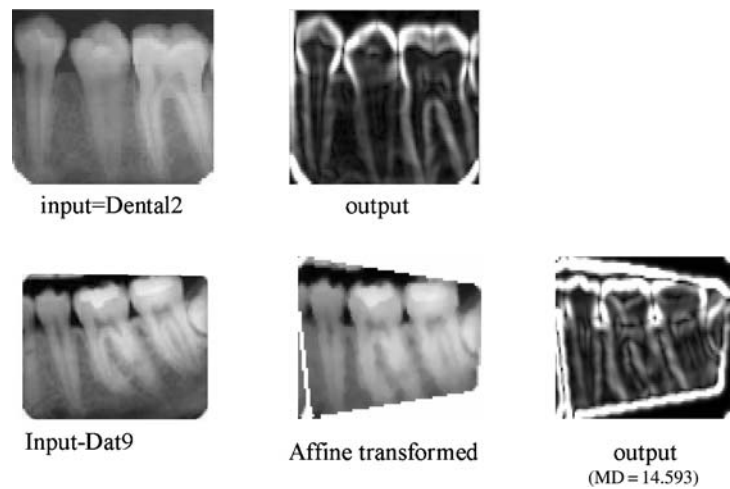
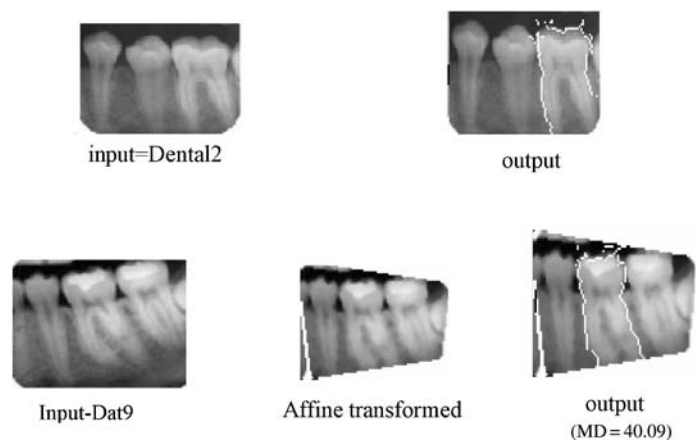
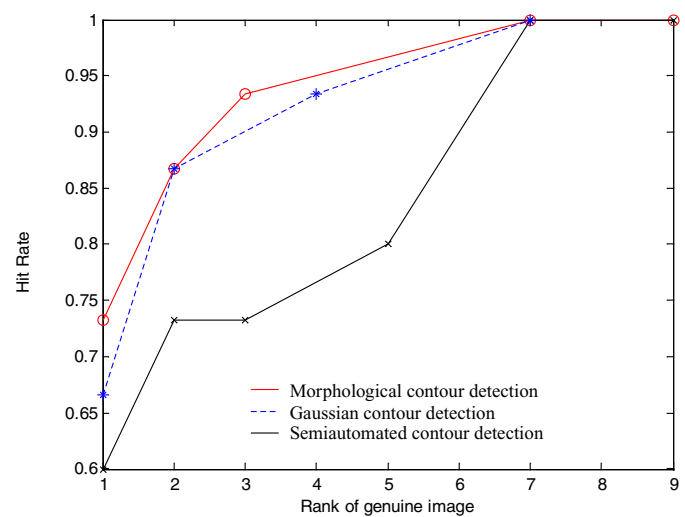
Similarly for Gaussian edge preserving filtering technique, the size of the Mask has been chosen as 7×7 for better contour extraction. The Kernel for X and Y directions were designed with the expression (12) and are represented as H_X and $\bar{H}_Y$. Since $\bar{H}_Y$ is the 90° rotated version of H_X , the rows and columns values were interchanged in the filter. The ' σ ' value has been chosen as 1. These filters were used to extract the boundary of the tooth.

In Gaussian filtering approach, out of 15 query images, 9 of them were ranked first as genuine images for the, 2 were among top 2 retrievals and 2 were among top 3 retrievals.

So, correspondingly the Hit rate is calculated for morphological contour detector, Gaussian filter and semi-automated method and plotted against the ranking of the Genuine teeth is shown in the Fig. 14. The parameters like Hit rate, Matching distance, computation time and Storage capacity are calculated and tabulated in Table 2.

Conclusion and future work

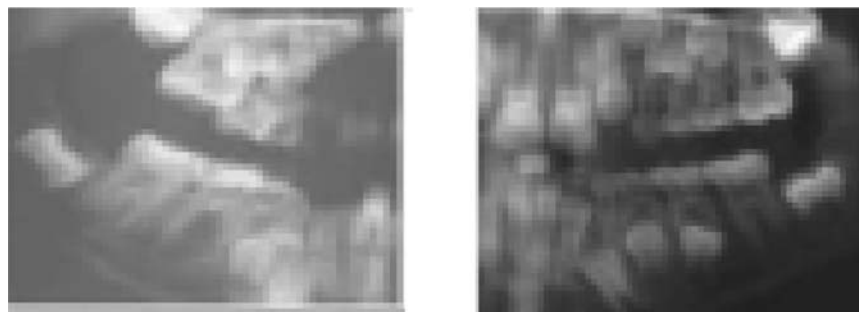
An automated dental image segmentation algorithm that handles bitewing and periapical dental images based on

Fig. 12 Results of Gaussian filtering**Fig. 13** Results of the semi automatic contour extraction method**Fig. 14** Comparative retrieval performance

mathematical morphology is proposed here. This method involves three stages: radiograph segmentation, tooth feature extraction, and tooth feature matching. After the teeth contours are extracted, the final decision is done using match-

ing distance. However, it is difficult to apply the proposed method in situations where the images are very blurred (Fig. 15), and the query shape is partially occluded so that there is not enough information.

Fig. 15 The images are too blurred for extraction



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References

1. Jain, A. K., Chen, H., and Minut, S., *Dental Biometrics: Human Identification Using Dental Radiographs*. Department of Computer Science, Michigan State University, East Lansing, MI, To appear in AVBPA 2003.
2. Jain, A. K., Chen, H., Matching of dental X-ray images for human identification. *Pattern Recognition—J. Patt. Recog. Soc.* 2003.
3. Jain, A., Bolle, R., and Pankanti, S., *Biotmetricsñ Personal Identification in Networked Society*. Kluwer Academic Publishers, 1999.
4. Han, S. P., A globally convergent method for nonlinear programming. *J. Optimiz. Theor. Appl.* 22:297, 1977.
5. Laganière, R., *Morphological Contour Detection*. School of Information Technology and Engineering University of Ottawa, Ottawa, Ontario Canada.
6. Yu, Z., *Recursive Implementation of Anisotropic Filters*. Department of Computer Science, University of Texas at Austin, TX, 1993.
7. deBoor, C., B(asic)-spline basics. In: L. Piegl (Ed.), *Fundamental Developments of Computer-Aided Geometric Modeling*. Academic Press, New York, 1993, pp. 27–49.
8. Jain, A. K., Chen, H., Minut, S., Dental biometrics: human identification using dental radiographs. In: *Proceedings of the Fourth International Conference on AVBPA*. Guildford, UK, 2003, pp. 429–437.
9. Cavalcanti, M., Ruprecht, A., Johnson, W., Thomas, S., Jakobsen, J., and Paulo, S., Oral and maxillofacial radiology. *Oral. Surg. Oral. Med. Oral. Pathol.* 88:353–357, 1999.
10. Pretty, I. A., and Sweet, D., A look at forensic dentistry—part 1: The role of teeth in the determination of human identity. *Br. Dental J.* 190(7):359–366, 2001.
11. J. Am. Dent. Assoc., American board of forensic odontology. *Body Identification Guidelines* 125:1244–1254, 1994.